

Modelling Volatility: The ARCH Model

Week 2 – Problem Set 1

FI 362 – Advanced Financial Econometrics | Shanghai Jiao Tong University

Instructions. Attempt all problems. Show *every* step of your working – in this course the derivation is worth more than the final number. Notation follows the lecture: returns are decomposed as $r_t = \mu_t + a_t$ with $a_t = \sigma_t \varepsilon_t$, where $\varepsilon_t \stackrel{iid}{\sim} (0, 1)$ are the standardised innovations, $\sigma_t^2 = \text{Var}(r_t | \mathcal{F}_{t-1})$ is the conditional variance, and $\mathcal{F}_{t-1} = \{a_{t-1}, a_{t-2}, \dots\}$ is the information set. Unless stated otherwise, take the mean to be zero ($\mu_t = 0$, so $r_t = a_t$). **Full step-by-step solutions are provided at the end of this problem set** – attempt each part before consulting them.

1. Concepts and intuition

(20 pts)

Answer each in two or three sentences.

- A1. Blind spot of linear models.** Explain precisely why an $\text{ARMA}(p, q)$ model cannot capture volatility clustering. Which moment does it model, and which does it assume constant?
- A2. Two variances.** Distinguish the *conditional* variance $\sigma_t^2 = \text{Var}(r_t | \mathcal{F}_{t-1})$ from the *unconditional* variance $\text{Var}(r_t)$. Which of the two does an ARCH model describe, and which is a single fixed number?
- A3. Uncorrelated $\neq$ independent.** Daily returns are serially *uncorrelated*, yet $|r_t|$ and r_t^2 are strongly autocorrelated for many lags. What kind of dependence does this reveal, what does it rule out, and what model class does it motivate?
- A4. Why square (or absolute-value) the returns?** Explain why r_t^2 and $|r_t|$ are used as “volatility proxies.” In what sense is r_t^2 a one-day estimate of σ_t^2 when $\mu_t = 0$?
- A5. Stylised facts.** State the three stylised facts of daily asset returns that motivate the ARCH model, and name the tool used to detect each.

2. Analytical properties of ARCH(1)

(35 pts)

Throughout this section let

$$a_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} (0, 1), \quad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2, \quad \alpha_0 > 0, \quad 0 \leq \alpha_1 < 1,$$

and assume the process is (weakly) stationary with the relevant moments finite.

- B1. Zero mean.** Show that $\mathbb{E}[a_t] = 0$. State clearly where you use that σ_t is $\mathcal{F}_{t-1}$ -measurable and that ε_t is independent of $\mathcal{F}_{t-1}$.
- B2. No autocorrelation.** Show that $\text{Cov}(a_t, a_{t-k}) = 0$ for every $k \geq 1$; i.e. $\{a_t\}$ is white noise. Why does this reconcile ARCH with the observed lack of mean predictability?

B3. Unconditional variance. Using the law of iterated expectations, derive

$$\text{Var}(a_t) = \frac{\alpha_0}{1 - \alpha_1}.$$

Explain why the condition $\alpha_1 < 1$ is exactly what makes this finite and positive.

B4. The squares are AR(1). Define $\eta_t = a_t^2 - \sigma_t^2$. Show that $\mathbb{E}[\eta_t | \mathcal{F}_{t-1}] = 0$, hence that

$$a_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \eta_t$$

is an AR(1) in a_t^2 . Deduce that its autocorrelation function is $\rho_k(a_t^2) = \alpha_1^k$. Interpret this as “volatility clustering.”

B5. Fat tails (harder). Assume $\varepsilon_t \sim \mathcal{N}(0, 1)$ (so $\mathbb{E}[\varepsilon_t^4] = 3$) and that the fourth moment is finite. Prove that the (unconditional) kurtosis of a_t is

$$\frac{\mathbb{E}[a_t^4]}{(\mathbb{E}[a_t^2])^2} = \frac{3(1 - \alpha_1^2)}{1 - 3\alpha_1^2} > 3,$$

and state the parameter restriction required for the fourth moment to exist.

3. Numerical computation and forecasting

(35 pts)

Problem C1 – a single-stock ARCH(1)

An analyst models a stock’s daily percentage returns ($\mu_t = 0$) as an ARCH(1):

$$\sigma_t^2 = 0.4 + 0.6 a_{t-1}^2 \quad (\text{units: } \%^2).$$

Yesterday’s return was $a_t = -3\%$.

- Verify the stationarity (finite-variance) condition.
- Compute the unconditional variance and the long-run daily volatility.
- Compute the one-day-ahead conditional variance σ_{t+1}^2 and volatility σ_{t+1} . Is tomorrow calmer or stormier than average? Why?
- Compute the two- and three-day-ahead variance forecasts $\sigma_t^2(2)$ and $\sigma_t^2(3)$.
- To what level do the h -day forecasts converge, and at what rate? Write the closed-form expression for $\sigma_t^2(h) - \sigma^2$.
- Give an approximate 95% interval for tomorrow’s return assuming Gaussian innovations.
- Careful:* does this model have finite kurtosis? Justify with the condition from B5, and interpret.

Problem C2 – an ARCH(2)

A second series is modelled as ARCH(2):

$$\sigma_t^2 = 0.2 + 0.3 a_{t-1}^2 + 0.2 a_{t-2}^2.$$

- State and check the stationarity condition, and compute the unconditional variance.
- The two most recent shocks are $a_t = 1.0$ and $a_{t-1} = 0.5$. Compute the one-day-ahead conditional variance σ_{t+1}^2 .
- Compute the two-day-ahead forecast $\sigma_t^2(2)$. *Hint:* which past squared shocks are still known at the forecast origin t , and which must be replaced by their conditional expectation?

4. Empirical exercise (optional, in R)

(10 pts)

Using the course `data/` folder and the script `Day1_ARCH_code.R` (S&P 500 daily returns, 2000–2026):

- D1.** Report the sample excess kurtosis of the returns and the Jarque–Bera p -value. What do they say about normality?
- D2.** Run the Ljung–Box test at lag 20 on r_t and on r_t^2 . Contrast the two $Q(20)$ statistics and explain what the contrast implies.
- D3.** Fit an ARCH(1) by maximum likelihood. Report $\hat{\alpha}_0, \hat{\alpha}_1$ and the implied long-run volatility. Then test the squared standardised residuals for remaining ARCH effects – is one lag enough? What does this motivate?